

Prostaglandin

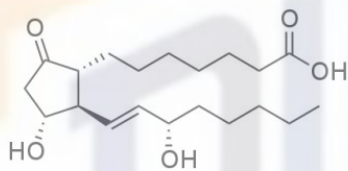
The **prostaglandins (PG)** are a group of physiologically active lipid compounds called eicosanoids^[1] having diverse hormone-like effects in animals.

Prostaglandins have been found in almost every tissue in humans and other animals.

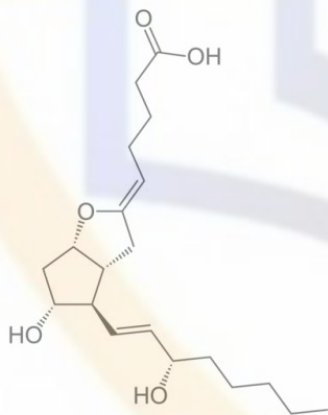
They are derived enzymatically from the fatty acid arachidonic acid.^[2] Every

prostaglandin contains 20 carbon atoms, including a 5-carbon ring. They are a

subclass of eicosanoids and of the prostanoid class of fatty acid derivatives.



*E*₁-Alprostadiil



*I*₂-Prostacyclin

The structural differences between prostaglandins account for their different biological activities. A given prostaglandin may have different and even opposite effects in different tissues in some cases. The ability of the same prostaglandin to stimulate a reaction in one tissue and inhibit the same reaction in another tissue is determined by the type of receptor to which the prostaglandin binds. They act as autocrine or paracrine factors with their target cells present in the immediate vicinity of the site of their secretion. Prostaglandins differ from endocrine hormones in that they are not produced at

a specific site but in many places throughout the human body.

Prostaglandins are powerful, locally-acting vasodilators and inhibit the aggregation of blood platelets. Through their role in vasodilation, prostaglandins are also involved in inflammation. They are synthesized in the walls of blood vessels and serve the physiological function of preventing needless clot formation, as well as regulating the contraction of smooth muscle tissue.^[3] Conversely, thromboxanes (produced by platelet cells) are vasoconstrictors and facilitate platelet

aggregation. Their name comes from their role in clot formation (thrombosis).

Specific prostaglandins are named with a letter (which indicates the type of ring structure) followed by a number (which indicates the number of double bonds in the hydrocarbon structure). For example, prostaglandin E1 is abbreviated PGE₁, and prostaglandin I2 is abbreviated PGI₂.

History and name

Systematic studies of prostaglandins began in 1930, when Kurzrock and Lieb found that human seminal fluid caused either stimulation or relaxation of strips of

isolated human uterus. They noted the curious finding that uteri from patients who had gone through successful pregnancies responded to the fluid with relaxation, while uteri from sterile women responded with contraction upon addition of this seminal fluid.^[4] The name *prostaglandin* derives from the prostate gland, chosen when prostaglandin was first isolated from seminal fluid in 1935 by the Swedish physiologist Ulf von Euler,^[5] and independently by the Irish-English physiologist Maurice Walter Goldblatt (1895–1967).^{[6][7][8]} Prostaglandins were believed to be part of the prostatic secretions, and eventually were discovered

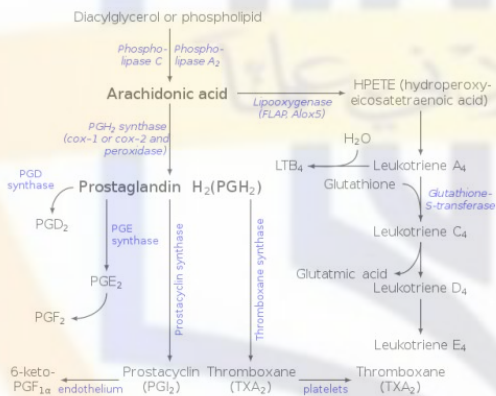
to be produced by the seminal vesicles. Later, it was shown that many other tissues secrete prostaglandins and that they perform a variety of functions. The first total syntheses of prostaglandin $F_{2\alpha}$ and prostaglandin E_2 were reported by E. J. Corey in 1969,^[9] an achievement for which he was awarded the Japan Prize in 1989.

In 1971, it was determined that aspirin-like drugs could inhibit the synthesis of prostaglandins. The biochemists Sune K. Bergström, Bengt I. Samuelsson and John R. Vane jointly received the 1982 Nobel

Prize in Physiology or Medicine for their research on prostaglandins.

Biochemistry

Biosynthesis



Biosynthesis of eicosanoids

Prostaglandins are found in most tissues and organs. They are produced by almost all nucleated cells. They are autocrine and

paracrine lipid mediators that act upon platelets, endothelium, uterine and mast cells. They are synthesized in the cell from the fatty acid arachidonic acid.^[2]

Arachidonic acid is created from diacylglycerol via phospholipase-A₂, then brought to either the cyclooxygenase pathway or the lipoxygenase pathway. The cyclooxygenase pathway produces thromboxane, prostacyclin and prostaglandin D, E and F. Alternatively, the lipoxygenase enzyme pathway is active in leukocytes and in macrophages and synthesizes leukotrienes.

Release of prostaglandins from the cell

Prostaglandins were originally believed to leave the cells via passive diffusion because of their high lipophilicity. The discovery of the prostaglandin transporter (PGT, SLCO2A1), which mediates the cellular uptake of prostaglandin, demonstrated that diffusion alone cannot explain the penetration of prostaglandin through the cellular membrane. The release of prostaglandin has now also been shown to be mediated by a specific transporter, namely the multidrug resistance protein 4 (MRP4, ABCC4), a

member of the ATP-binding cassette transporter superfamily. Whether MRP4 is the only transporter releasing prostaglandins from the cells is still unclear.

Cyclooxygenases

Prostaglandins are produced following the sequential oxygenation of arachidonic acid, DGLA or EPA by cyclooxygenases (COX-1 and COX-2) and terminal prostaglandin synthases. The classic dogma is as follows:

- COX-1 is responsible for the baseline levels of prostaglandins.

- COX-2 produces prostaglandins through stimulation.

However, while COX-1 and COX-2 are both located in the blood vessels, stomach and the kidneys, prostaglandin levels are increased by COX-2 in scenarios of inflammation and growth.

Prostaglandin E synthase

Prostaglandin E₂ (PGE₂) — the most abundant prostaglandin^[10] — is generated from the action of prostaglandin E synthases on prostaglandin H₂ (prostaglandin H₂, PGH₂). Several prostaglandin E synthases have been

identified. To date, microsomal prostaglandin E synthase-1 emerges as a key enzyme in the formation of PGE_2 .

Other terminal prostaglandin synthases

Terminal prostaglandin synthases have been identified that are responsible for the formation of other prostaglandins. For example, hematopoietic and lipocalin prostaglandin D synthases (hPGDS and lPGDS) are responsible for the formation of PGD_2 from PGH_2 . Similarly, prostacyclin (PGI_2) synthase (PGIS) converts PGH_2 into PGI_2 . A thromboxane synthase (TxAS) has also been identified. Prostaglandin-F

synthase (PGFS) catalyzes the formation of $9\alpha,11\beta$ -PGF_{2 α,β} from PGD₂ and PGF_{2 α} from PGH₂ in the presence of NADPH. This enzyme has recently been crystallized in complex with PGD₂^[11] and bimatoprost^[12] (a synthetic analogue of PGF_{2 α}).

Functions

There are currently ten known prostaglandin receptors on various cell types. Prostaglandins ligate a sub-family of cell surface seven-transmembrane receptors, G-protein-coupled receptors. These receptors are termed DP1-2, EP1-4, FP, IP1-2, and TP, corresponding to the receptor that ligates the corresponding

prostaglandin (e.g., DP1-2 receptors bind to PGD2).

The diversity of receptors means that prostaglandins act on an array of cells and have a wide variety of effects such as:

- create eicosanoids hormones
- acts on thermoregulatory center of hypothalamus to produce fever
- increases mating behaviors in goldfish^[13]
- Prostaglandins are released during menstruation, due to the destruction of the endometrial cells, and the resultant release of their contents.^[14] Release of

prostaglandins and other inflammatory mediators in the uterus cause the uterus to contract. These substances are thought to be a major factor in primary dysmenorrhea.^{[15][16][17]}

Types

The following is a comparison of different types of prostaglandin, including prostaglandin I₂ (prostacyclin; PGI₂), prostaglandin D₂ (PGD₂), prostaglandin E₂ (PGE₂), and prostaglandin F_{2α} (PGF_{2α}).^[18]

Type	Receptor	Receptor type	Function
<u>PGI₂</u>	<u>IP</u>	<u>G_s</u>	• <u>vasodilation</u> • inhibit <u>platelet aggregation</u> • <u>bronchodilation</u>
<u>PGD₂</u>	<u>PTGDR (DP1) and CRT2 (DP2)</u>	<u>GPCR</u>	• produced by mast cells; recruits Th2 cells, eosinophils, and basophils • In <u>mammalian</u> organs, large amounts of PGD2 are found only in the brain and in mast cells • Critical to development of allergic diseases such as asthma
<u>PGE₂</u>	<u>EP₁</u>	<u>G_q</u>	• <u>bronchoconstriction</u> • <u>GI tract smooth muscle</u> contraction
	<u>EP₂</u>	<u>G_s</u>	• <u>bronchodilation</u> • <u>GI tract smooth muscle</u> relaxation • <u>vasodilation</u>
	<u>EP₃</u>	<u>G_i</u>	• ↓ <u>gastric acid</u> secretion • ↑ <u>gastric mucus</u> secretion • <u>uterus</u> contraction (when pregnant) • <u>GI tract smooth muscle</u> contraction • <u>lipolysis</u> inhibition • ↑ <u>autonomic neurotransmitters</u>^[19] • ↑ platelet response to their agonists^[20] and ↑ atherothrombosis in vivo^[21]
	Unspecified		• <u>hyperalgesia</u>^[19] • <u>pyrogenic</u>
<u>PGF_{2α}</u>	<u>FP</u>	<u>G_q</u>	• <u>uterus</u> contraction • <u>bronchoconstriction</u>

- | | | |
|--|--|---|
| | | <ul style="list-style-type: none">• urinary bladder contractions^[22]• vasoconstriction in cerebral circulation^[23] |
|--|--|---|

Role in pharmacology

Inhibition

Examples of prostaglandin antagonists are:

- NSAIDs (inhibit cyclooxygenase) and COX-2 selective inhibitors or coxibs
- Corticosteroids (inhibit phospholipase A2 production)
- Cyclopentenone prostaglandins may play a role in inhibiting inflammation

Clinical uses

Synthetic prostaglandins are used:

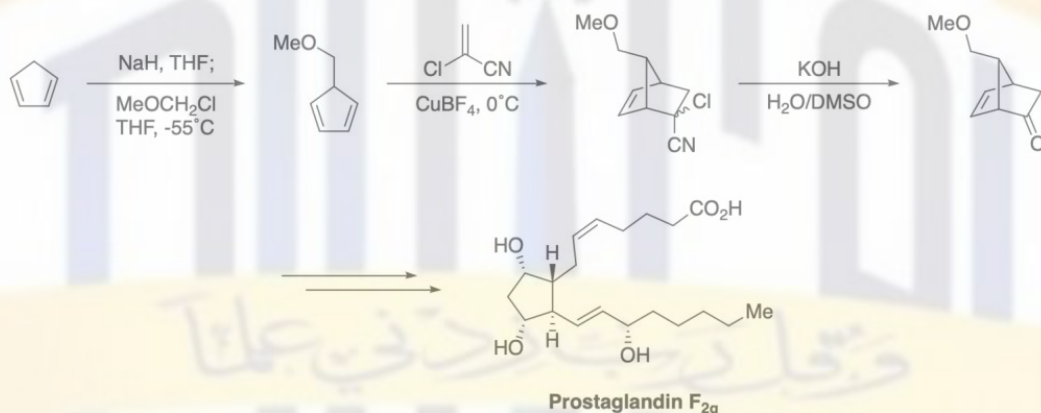
- To induce childbirth (parturition) or abortion (PGE_2 or PGF_2 , with or without mifepristone, a progesterone antagonist)
 - Induction of labour^[24]
- To prevent closure of ductus arteriosus in newborns with particular cyanotic heart defects (PGE_1)
- As a vasodilator in severe Raynaud syndrome or ischemia of a limb
- In pulmonary hypertension

- In treatment of glaucoma (as in bimatoprost ophthalmic solution, a synthetic prostamide analog with ocular hypotensive activity) (PGF_{2α})
- To treat erectile dysfunction or in penile rehabilitation following surgery (PGE₁ as alprostadil).^[25]
- To measure erect penis size in a clinical environment^[26]
- To treat egg binding in small birds^[27]

Synthesis

The original synthesis of prostaglandins F_{2α} and E₂ is shown below. It involves a Diels–Alder reaction which establishes the

relative stereochemistry of three contiguous stereocenters on the prostaglandin cyclopentane core.^[28]



Prostaglandin stimulants

Cold exposure and IUDs may increase prostaglandin production.^[29]

PUACP